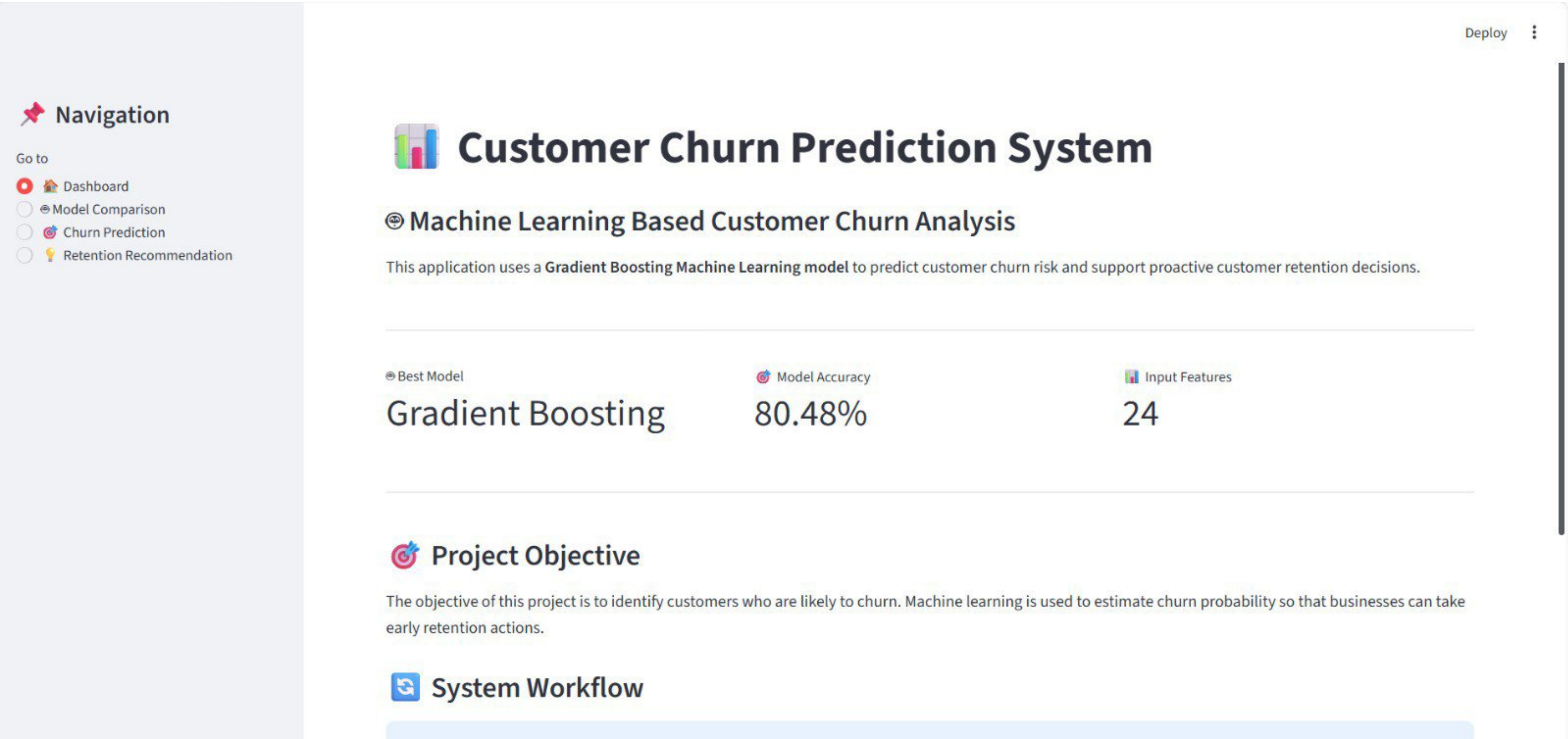

Deploy
📌 Navigation
Go to
🏠 Dashboard
Model Comparison
🎯 Churn Prediction
💡 Retention Recommendation
📊 Customer Churn Prediction System
Machine Learning Based Customer Churn Analysis
This application uses a Gradient Boosting Machine Learning model to predict customer churn risk and support proactive customer retention decisions.
Best Model
Gradient Boosting
🎯 Model Accuracy
80.48%
📊 Input Features
24
🎯 Project Objective
The objective of this project is to identify customers who are likely to churn. Machine learning is used to estimate churn probability so that businesses can take early retention actions.
🔄 System Workflow

Deploy

Best Model

Gradient Boosting

Model Accuracy

80.48%

Input Features

24

🎯 Project Objective

The objective of this project is to identify customers who are likely to churn. Machine learning is used to estimate churn probability so that businesses can take early retention actions.

🔄 System Workflow

Customer Data → Data Preprocessing → Gradient Boosting Model → Churn Probability → Risk Classification → Retention Recommendation

---

Customer Churn Prediction System | Gradient Boosting | Streamlit

Deploy

📌 Navigation

Go to

- 🏠 Dashboard
- 🤖 Model Comparison
- 🎯 Churn Prediction
- 💡 Retention Recommendation

🤖 Machine Learning Model Comparison

🏆 Best Model: Gradient Boosting | Accuracy: 80.48%

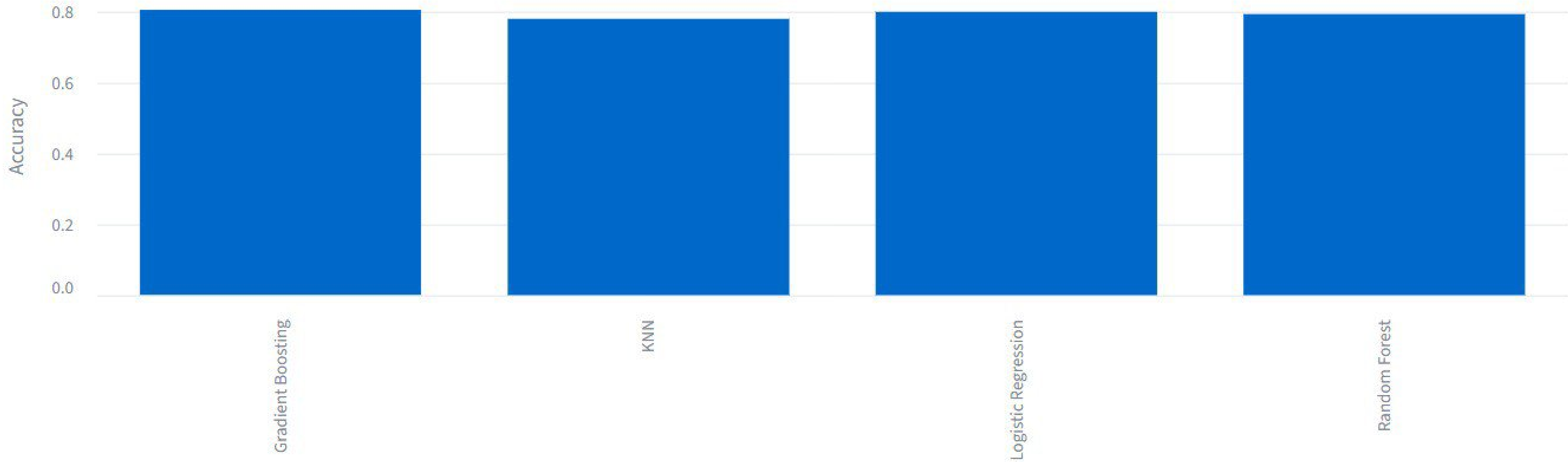
📋 Model Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	RMSE	R^2
Gradient Boosting	80.48%	0.6787	0.5027	0.5776	0.8463	0.3678	0.3061
Logistic Regression	79.91%	0.6542	0.516	0.577	0.8411	0.3723	0.2891
Random Forest	79.28%	0.6464	0.484	0.5535	0.8305	0.3766	0.2724
KNN	77.93%	0.5897	0.5535	0.571	0.8055	0.3925	0.2098

📈 Model Accuracy Comparison

Deploy

📈 Model Accuracy Comparison



Deploy

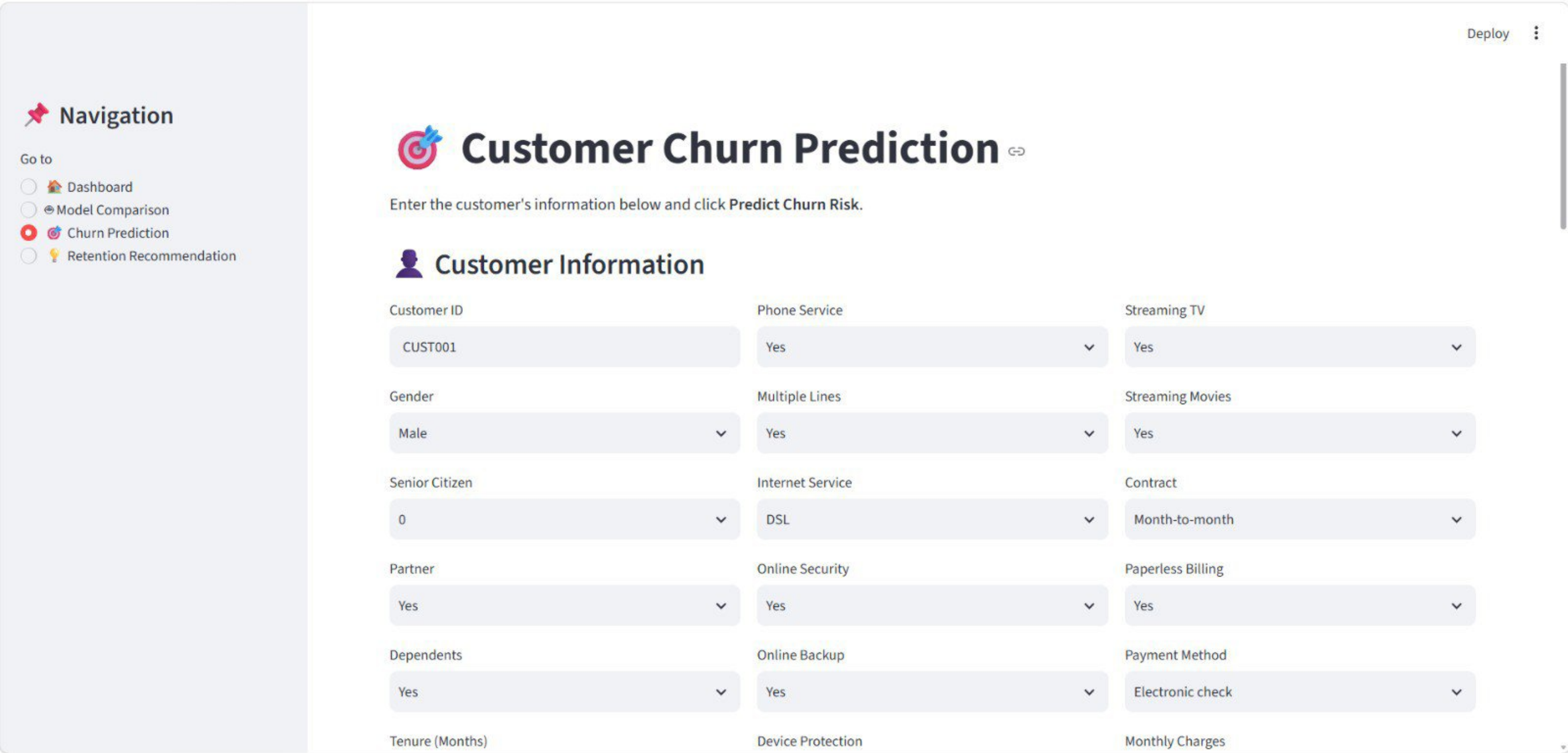
🏅 Model Ranking

Rank	Model	Accuracy
1	Gradient Boosting	80.48%
2	Logistic Regression	79.91%
3	Random Forest	79.28%
4	KNN	77.93%

🎯 Model Selection Conclusion

Gradient Boosting achieved the highest accuracy of **80.48%** among the evaluated models.

Therefore, **Gradient Boosting** was selected as the best-performing model for the Customer Churn Prediction system.


Deploy
Navigation
Go to
Dashboard
Model Comparison
Churn Prediction
Retention Recommendation
Customer Churn Prediction
Enter the customer's information below and click Predict Churn Risk.
Customer Information
Customer ID
CUST001
Phone Service
Yes
Streaming TV
Yes
Gender
Male
Multiple Lines
Yes
Streaming Movies
Yes
Senior Citizen
0
Internet Service
DSL
Contract
Month-to-month
Partner
Yes
Online Security
Yes
Paperless Billing
Yes
Dependents
Yes
Online Backup
Yes
Payment Method
Electronic check
Tenure (Months)
Device Protection
Monthly Charges

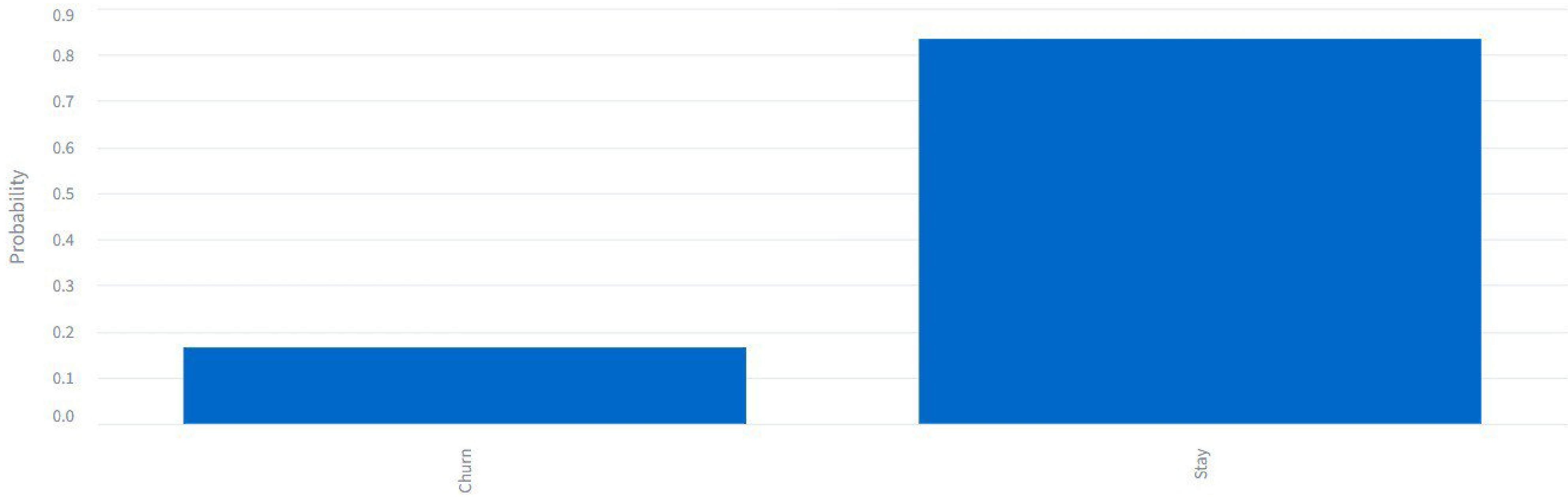
Deploy

Predict Churn Risk

📊 Prediction Result

Churn Probability	Risk Level	Prediction
16.56%	LOW	Stay

📈 Stay vs Churn Probability



Deploy ⋮

Churn

Stay

🟢 **Stay Probability:** 83.44%

🔴 **Churn Probability:** 16.56%

🚦 Churn Risk Assessment

🟢 LOW CHURN RISK

This customer has a low probability of churn.

💡 Retention Recommendation

Recommended actions:

- ⭐ Maintain excellent service quality
- 🎁 Provide loyalty rewards
- 📢 Introduce relevant services
- 🤝 Encourage a long-term relationship

Deploy

This customer has a low probability of churn.

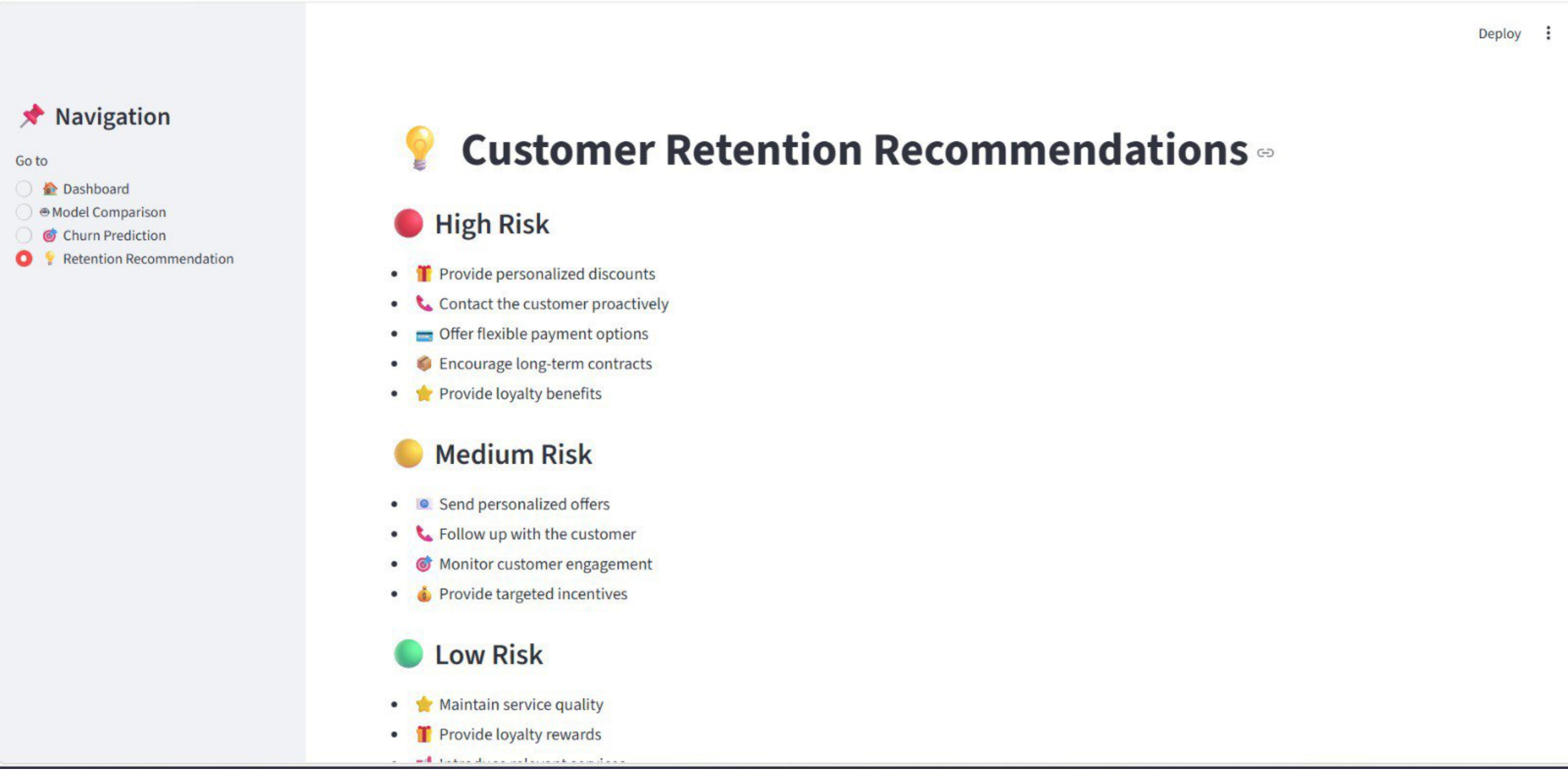
💡 Retention Recommendation

Recommended actions:

- ⭐ Maintain excellent service quality
- 🎁 Provide loyalty rewards
- 📢 Introduce relevant services
- 🤝 Encourage a long-term relationship

📋 Customer Prediction Summary

Customer ID	Prediction	Churn Probability	Stay Probability	Risk Level
CUST001	Stay	16.56%	83.44%	LOW


Deploy
Navigation
Go to
Dashboard
Model Comparison
Churn Prediction
Retention Recommendation
Customer Retention Recommendations
High Risk
Provide personalized discounts
Contact the customer proactively
Offer flexible payment options
Encourage long-term contracts
Provide loyalty benefits
Medium Risk
Send personalized offers
Follow up with the customer
Monitor customer engagement
Provide targeted incentives
Low Risk
Maintain service quality
Provide loyalty rewards

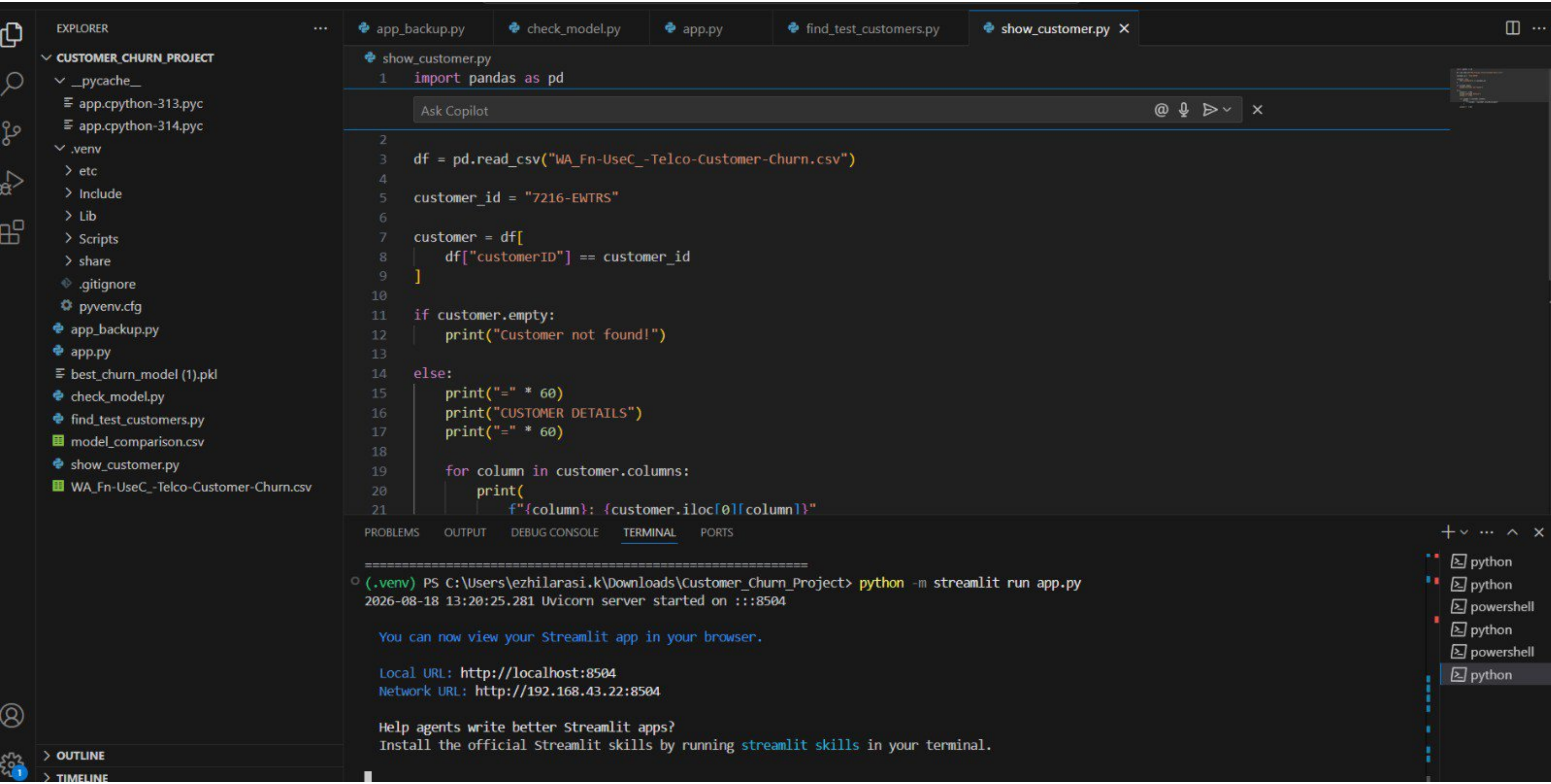
Deploy

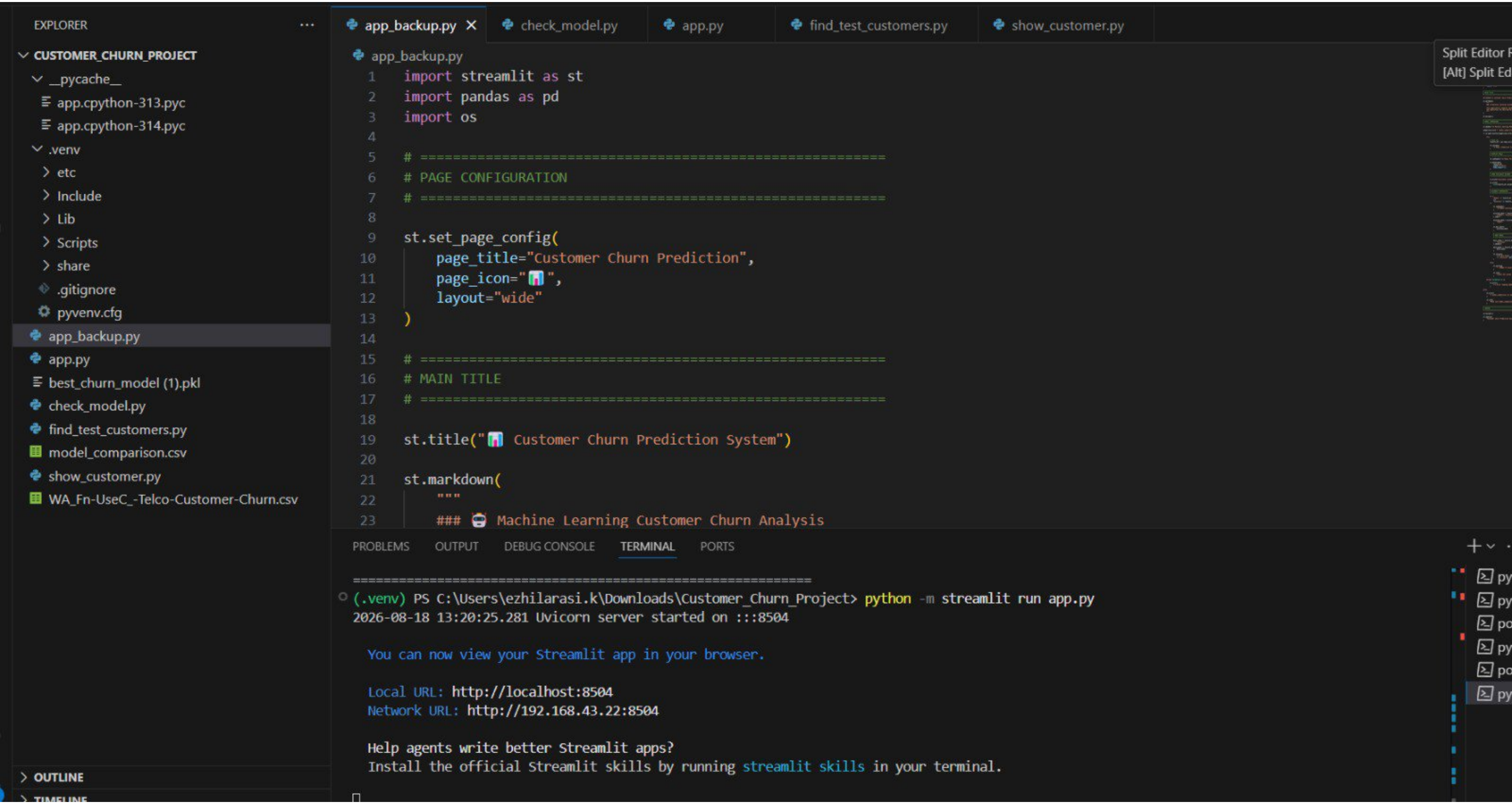
- 📞 Follow up with the customer
- 🎯 Monitor customer engagement
- 💰 Provide targeted incentives

🟢 Low Risk

- ⭐ Maintain service quality
- 🎁 Provide loyalty rewards
- 📢 Introduce relevant services
- 🤝 Encourage long-term relationships

🎯 Goal: Identify high-risk customers early and take proactive retention actions.


EXPLORER
CUSTOMER_CHURN_PROJECT
__pycache__
app.cpython-313.pyc
app.cpython-314.pyc
.venv
etc
Include
Lib
Scripts
share
.gitignore
pyvenv.cfg
app_backup.py
app.py
best_churn_model (1).pkl
check_model.py
find_test_customers.py
model_comparison.csv
show_customer.py
WA_Fn-UseC_-Telco-Customer-Churn.csv
OUTLINE
TIMELINE
app_backup.py
check_model.py
app.py
find_test_customers.py
show_customer.py
Ask Copilot
import pandas as pd
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
customer_id = "7216-EWTRS"
customer = df[
df["customerID"] == customer_id
]
if customer.empty:
print("Customer not found!")
else:
print("=" * 60)
print("CUSTOMER DETAILS")
print("=" * 60)
for column in customer.columns:
print(
f"{column}: {customer.iloc[0][column]}"
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
==
(.venv) PS C:\Users\ezhilarasi.k\Downloads\Customer_Churn_Project> python -m streamlit run app.py
2026-08-18 13:20:25.281 Uvicorn server started on :::8504
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8504
Network URL: http://192.168.43.22:8504
Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.
python
python
powershell
python
powershell
python


EXPLORER
CUSTOMER_CHURN_PROJECT
__pycache__
app.cpython-313.pyc
app.cpython-314.pyc
.venv
etc
Include
Lib
Scripts
share
.gitignore
pyvenv.cfg
app_backup.py
app.py
best_churn_model (1).pkl
check_model.py
find_test_customers.py
model_comparison.csv
show_customer.py
WA_Fn-UseC_-Telco-Customer-Churn.csv
OUTLINE
TIMELINE
app_backup.py
check_model.py
app.py
find_test_customers.py
show_customer.py
app_backup.py
import streamlit as st
import pandas as pd
import os
# ==
# PAGE CONFIGURATION
# ==
st.set_page_config(
page_title="Customer Churn Prediction",
page_icon="📊",
layout="wide"
)
# ==
# MAIN TITLE
# ==
st.title("📊 Customer Churn Prediction System")
st.markdown(
"""
🤖 Machine Learning Customer Churn Analysis
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
(.venv) PS C:\Users\ezhilarasi.k\Downloads\Customer_Churn_Project> python -m streamlit run app.py
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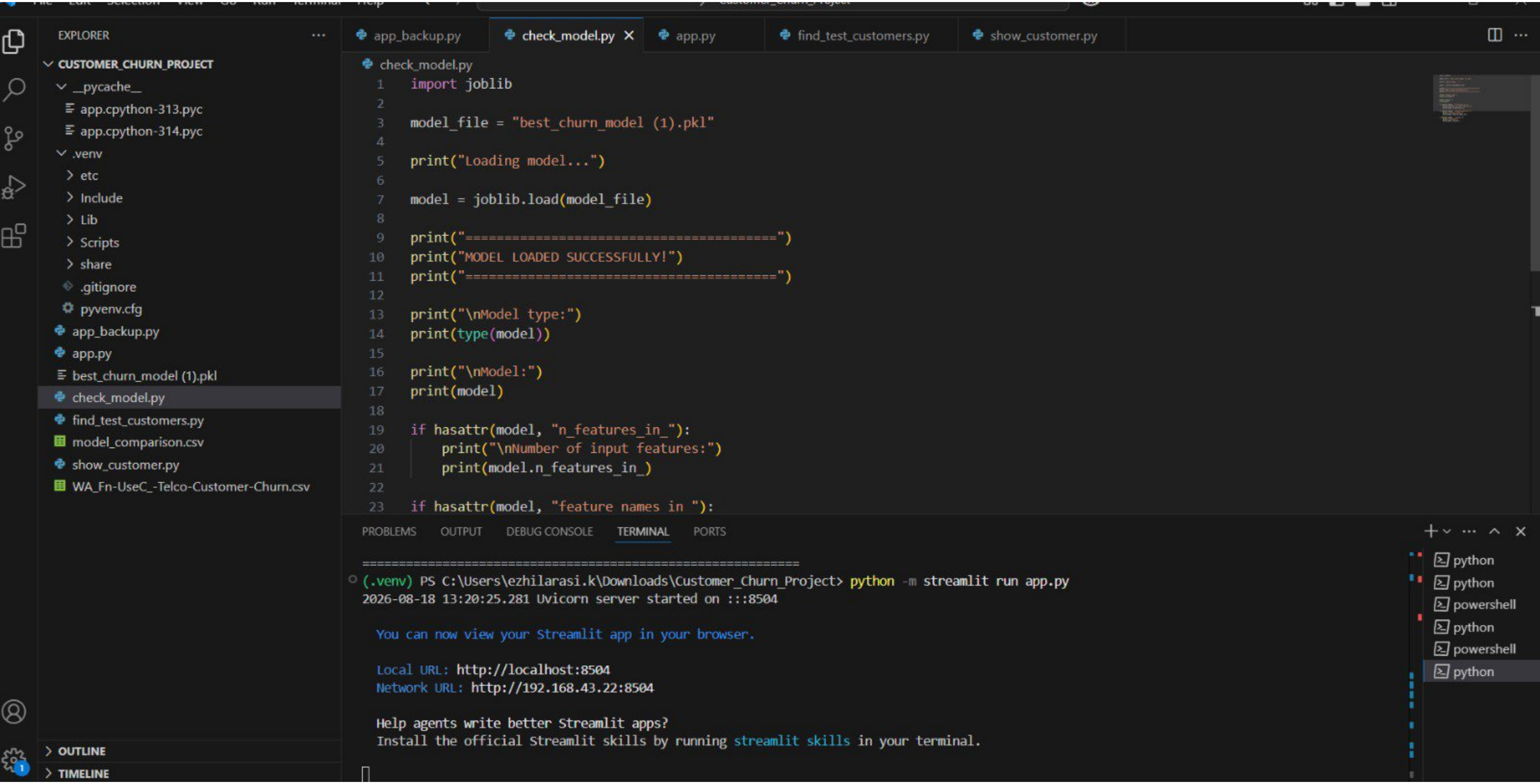
Split Editor R
[Alt] Split Ed


EXPLORER
CUSTOMER_CHURN_PROJECT
__pycache__
app.cpython-313.pyc
app.cpython-314.pyc
.venv
etc
Include
Lib
Scripts
share
.gitignore
pyvenv.cfg
app_backup.py
app.py
best_churn_model (1).pkl
check_model.py
find_test_customers.py
model_comparison.csv
show_customer.py
WA_Fn-UseC_-Telco-Customer-Churn.csv
OUTLINE
TIMELINE
app_backup.py
check_model.py
app.py
find_test_customers.py
show_customer.py
import joblib
model_file = "best_churn_model (1).pkl"
print("Loading model...")
model = joblib.load(model_file)
print("=======================================")
print("MODEL LOADED SUCCESSFULLY!")
print("=======================================")
print("\nModel type:")
print(type(model))
print("\nModel:")
print(model)
if hasattr(model, "n_features_in_"):
print("\nNumber of input features:")
print(model.n_features_in_)
if hasattr(model, "feature_names_in_"):
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
(.venv) PS C:\Users\ezhilarasi.k\Downloads\Customer_Churn_Project> python -m streamlit run app.py
2026-08-18 13:20:25.281 Uvicorn server started on :::8504
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8504
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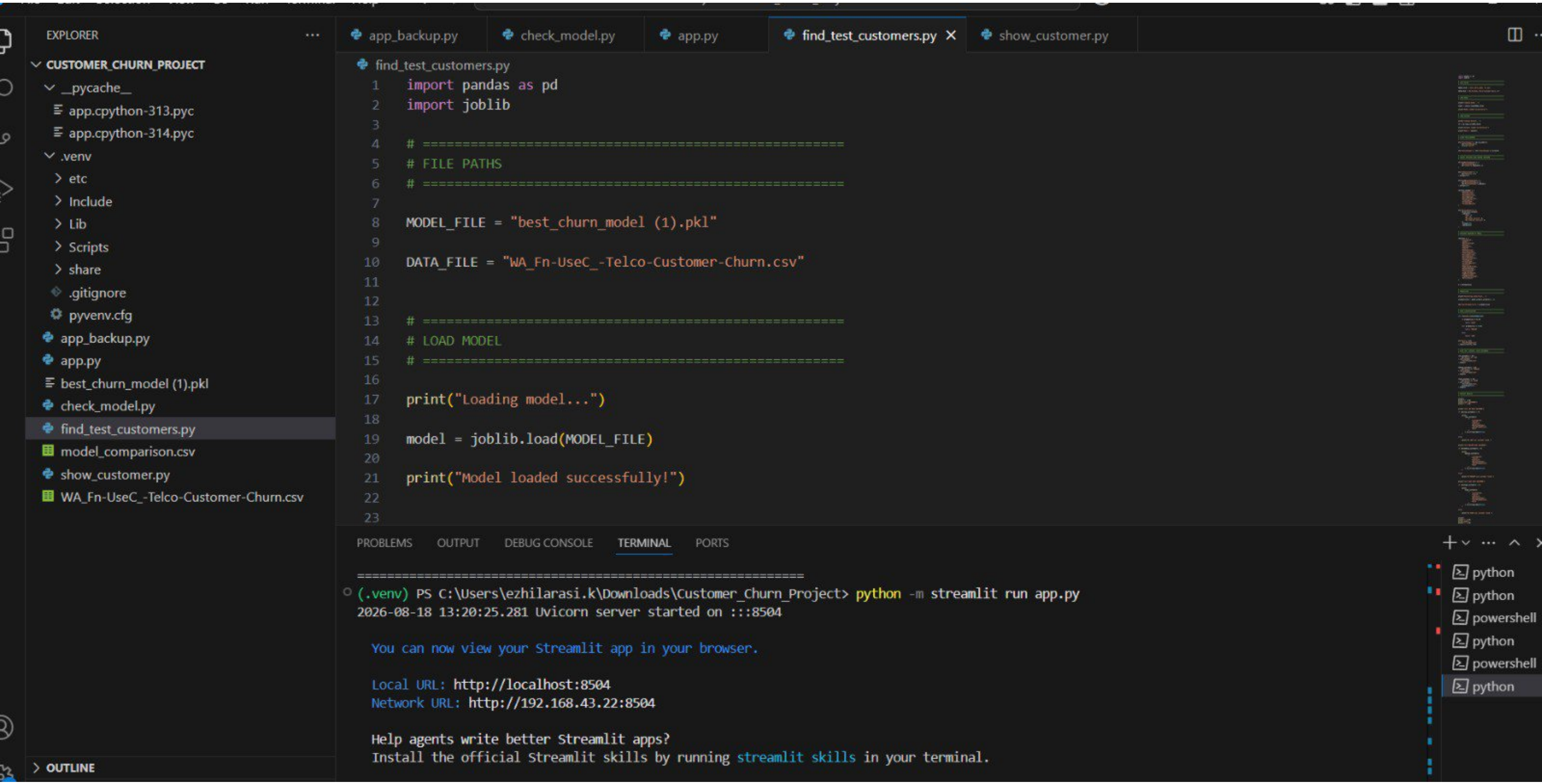
Install the official Streamlit skills by running streamlit skills in your terminal.
python
python
powershell
python
powershell
python


EXPLORER
CUSTOMER_CHURN_PROJECT
__pycache__
app.cpython-313.pyc
app.cpython-314.pyc
.venv
etc
Include
Lib
Scripts
share
.gitignore
pyvenv.cfg
app_backup.py
app.py
best_churn_model (1).pkl
check_model.py
find_test_customers.py
model_comparison.csv
show_customer.py
WA_Fn-UseC_-Telco-Customer-Churn.csv
OUTLINE
app_backup.py
check_model.py
app.py
find_test_customers.py
show_customer.py
find_test_customers.py
import pandas as pd
import joblib
# ==
# FILE PATHS
# ==
MODEL_FILE = "best_churn_model (1).pkl"
DATA_FILE = "WA_Fn-UseC_-Telco-Customer-Churn.csv"
# ==
# LOAD MODEL
# ==
print("Loading model...")
model = joblib.load(MODEL_FILE)
print("Model loaded successfully!")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
(.venv) PS C:\Users\ezhilarasi.k\Downloads\Customer_Churn_Project> python -m streamlit run app.py
2026-08-18 13:20:25.281 Uvicorn server started on :::8504
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8504
Network URL: http://192.168.43.22:8504
Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.
python
python
powershell
python
powershell
python

Customer Churn Prediction & Risk Analytics System.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM
Disk

[ ]

```
# ============================================================
# CUSTOMER CHURN PREDICTION & RISK ANALYTICS SYSTEM
# Advanced Machine Learning Project
# ============================================================

print("Customer Churn Prediction & Risk Analytics System")
print("Machine Learning | Model Comparison | Explainable AI")
```

```
Customer Churn Prediction & Risk Analytics System
Machine Learning | Model Comparison | Explainable AI
```

[ ]

```
!pip install -q pandas numpy matplotlib seaborn scikit-learn \
xgboost shap joblib streamlit
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.5/10.5 MB 81.7 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.4/11.4 MB 103.5 MB/s eta 0:00:00
```

[ ]

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import (
    train_test_split,
    StratifiedKFold,
    cross_val_score,
```

```python
from sklearn.model_selection import (
    train_test_split,
    StratifiedKFold,
    cross_val_score,
    GridSearchCV
)

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import (
    StandardScaler,
    OneHotEncoder
)

from sklearn.impute import SimpleImputer

from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import (
    RandomForestClassifier,
    GradientBoostingClassifier
)

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
```

```
    r2_score,
    roc_curve
)

import joblib
import warnings

warnings.filterwarnings("ignore")

sns.set_theme(style="whitegrid")
```

```
from google.colab import files

uploaded = files.upload()

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("Dataset Shape:", df.shape)

display(df.head())
```

Choose Files WA_Fn-Us...er-Churn.csv

WA_Fn-UseC_-Telco-Customer-Churn.csv(text/csv) - 977501 bytes, last modified: 8/18/2026 - 100% done

```
Saving WA_Fn-UseC_-Telco-Customer-Churn.csv to WA_Fn-UseC_-Telco-Customer-Churn.csv
Dataset Shape: (7043, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingT\
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	Nc
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	Nc
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	Nc
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	Nc
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	Nc

5 rows × 21 columns

```python
print("Dataset Information")
print("="*60)

print(df.info())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

print("\nStatistical Summary")
display(df.describe(include="all").T)
```

```
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

Duplicate Rows: 0
```

```
Duplicate Rows: 0

Statistical Summary
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
customerID	7043	7043	3186-AJIEK	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
gender	7043	2	Male	3555	NaN	NaN	NaN	NaN	NaN	NaN	NaN
SeniorCitizen	7043.0	NaN	NaN	NaN	0.162147	0.368612	0.0	0.0	0.0	0.0	1.0
Partner	7043	2	No	3641	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Dependents	7043	2	No	4933	NaN	NaN	NaN	NaN	NaN	NaN	NaN
tenure	7043.0	NaN	NaN	NaN	32.371149	24.559481	0.0	9.0	29.0	55.0	72.0
PhoneService	7043	2	Yes	6361	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MultipleLines	7043	3	No	3390	NaN	NaN	NaN	NaN	NaN	NaN	NaN
InternetService	7043	3	Fiber optic	3096	NaN	NaN	NaN	NaN	NaN	NaN	NaN
OnlineSecurity	7043	3	No	3498	NaN	NaN	NaN	NaN	NaN	NaN	NaN
OnlineBackup	7043	3	No	3088	NaN	NaN	NaN	NaN	NaN	NaN	NaN
DeviceProtection	7043	3	No	3095	NaN	NaN	NaN	NaN	NaN	NaN	NaN
TechSupport	7043	3	No	3473	NaN	NaN	NaN	NaN	NaN	NaN	NaN
StreamingTV	7043	3	No	2810	NaN	NaN	NaN	NaN	NaN	NaN	NaN
StreamingMovies	7043	3	No	2785	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
# Convert TotalCharges to numeric
df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"],
    errors="coerce"
)

# Check missing values
print(df.isnull().sum())

# Remove duplicates
df = df.drop_duplicates()

print("Final Shape:", df.shape)
```

```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
```

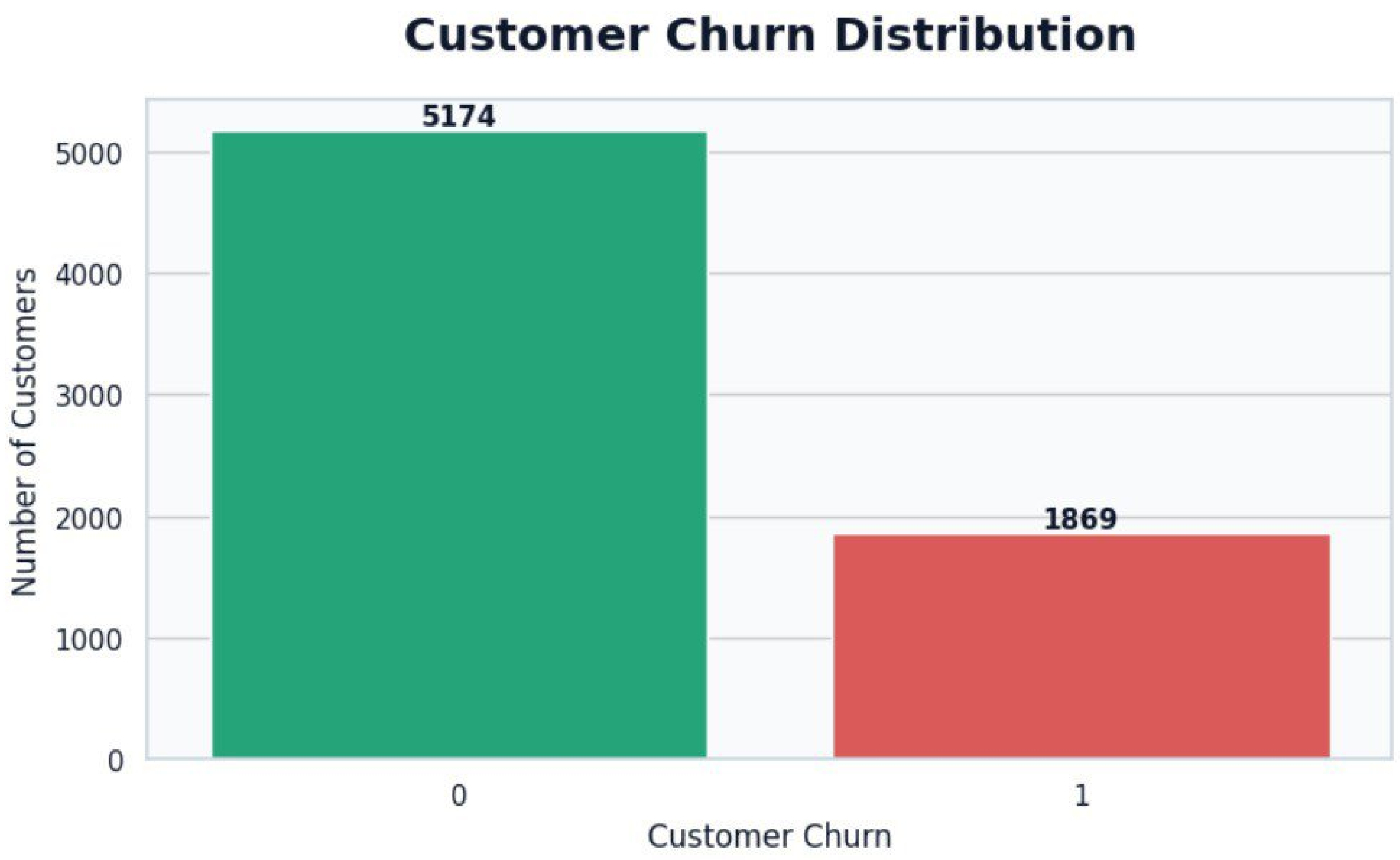
```python
df["Churn"] = df["Churn"].map({
    "Yes": 1,
    "No": 0
})

print(df["Churn"].value_counts())
```

```
Churn
0    5174
1    1869
Name: count, dtype: int64
```

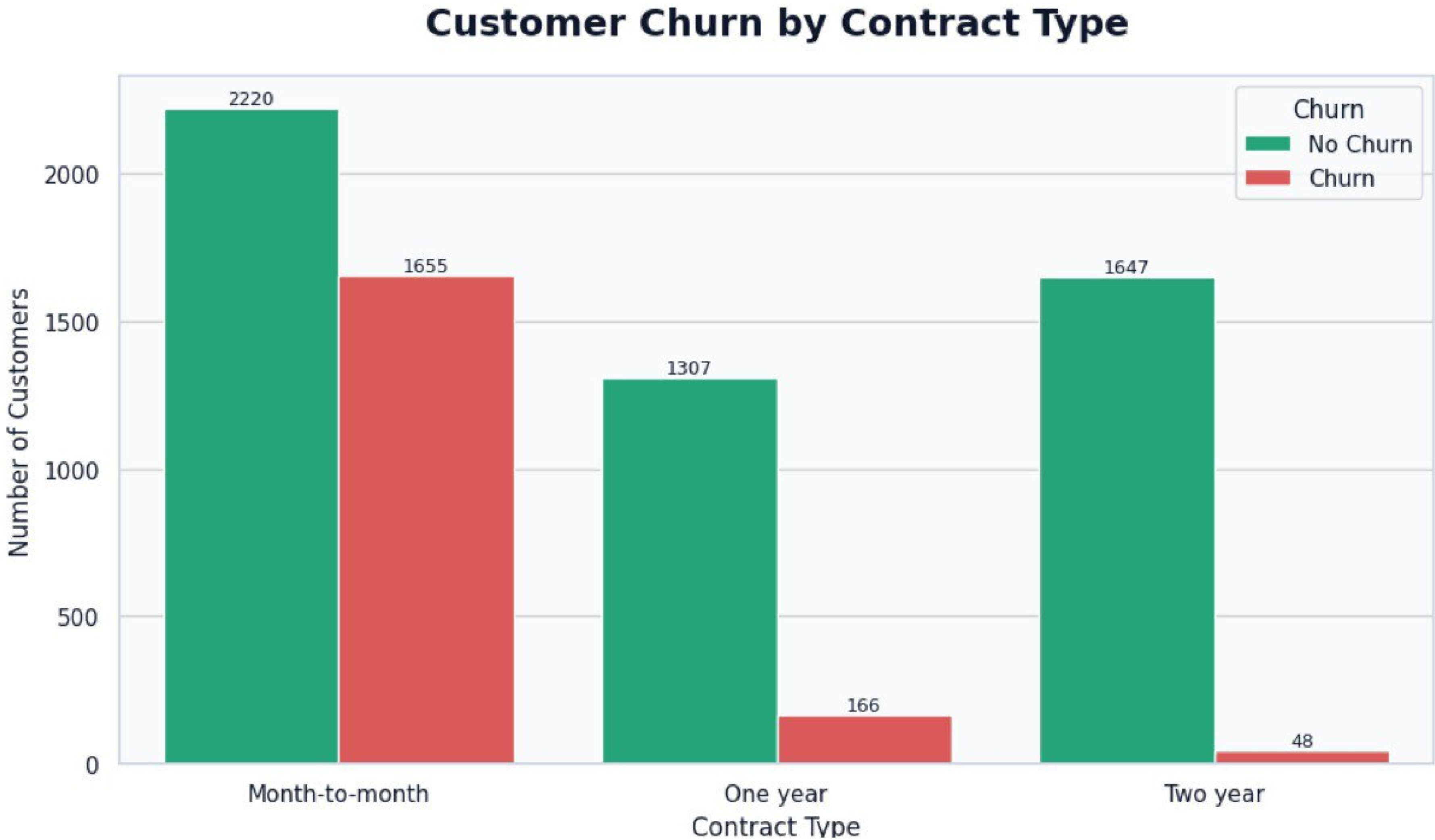
```python
# ============================================================
# 🎨 PROJECT VISUALIZATION THEME
# ============================================================

import matplotlib.pyplot as plt
import seaborn as sns

# Professional project colors
PRIMARY = "#2563EB"       # Blue
SECONDARY = "#06B6D4"     # Cyan
SUCCESS = "#10B981"       # Green
WARNING = "#F59E0B"       # Orange
DANGER = "#EF4444"        # Red
PURPLE = "#8B5CF6"        # Purple
```

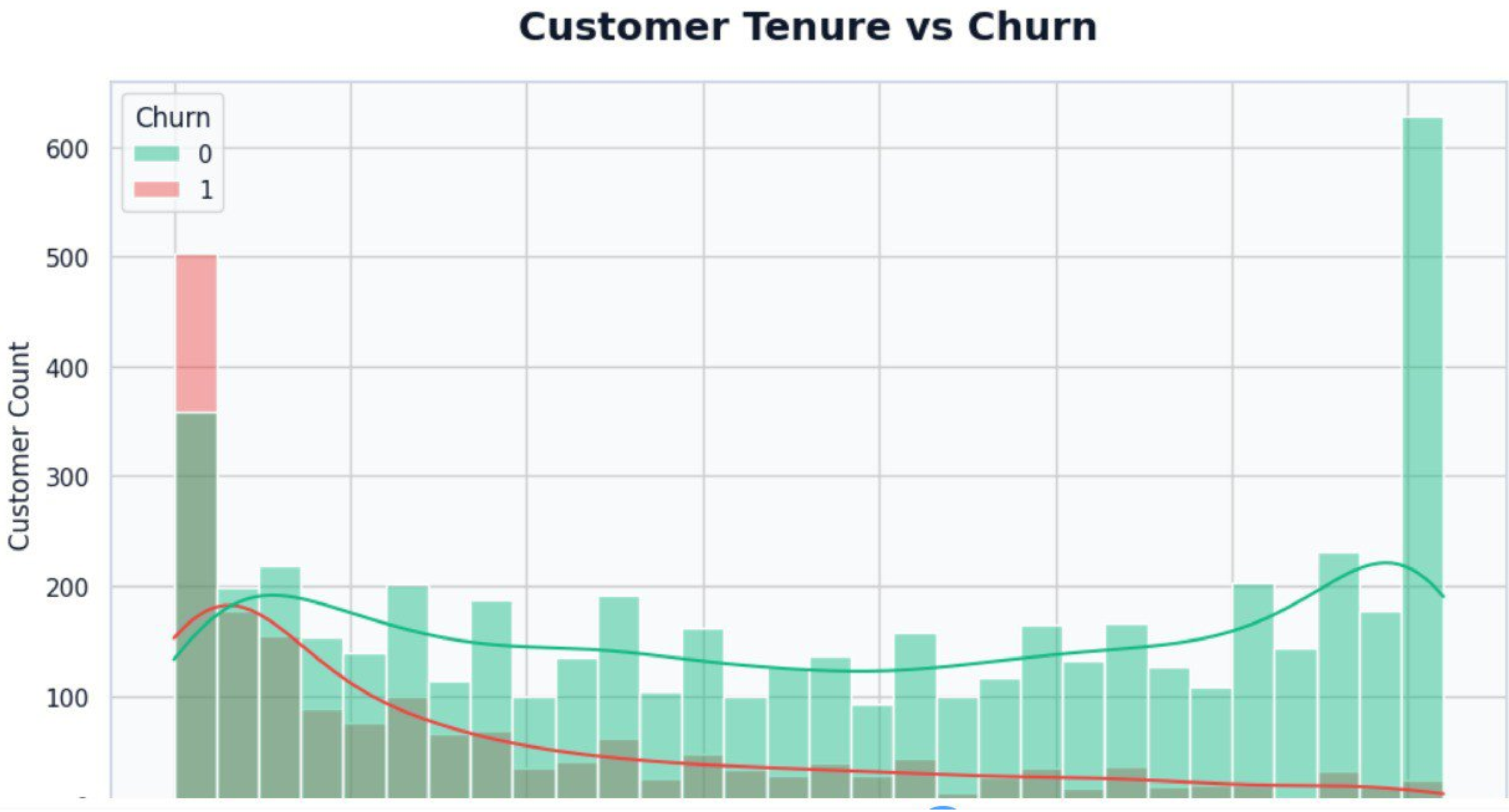
```python
# ============================================================
# 🎨 PROJECT VISUALIZATION THEME
# ============================================================

import matplotlib.pyplot as plt
import seaborn as sns

# Professional project colors
PRIMARY = "#2563EB"        # Blue
SECONDARY = "#06B6D4"      # Cyan
SUCCESS = "#10B981"        # Green
WARNING = "#F59E0B"        # Orange
DANGER = "#EF4444"         # Red
PURPLE = "#8B5CF6"         # Purple
DARK = "#0F172A"           # Dark Navy
LIGHT = "#F8FAFC"          # Light Background
GRAY = "#64748B"           # Gray

# Model colors
MODEL_COLORS = {
    "Logistic Regression": "#2563EB",
    "KNN": "#8B5CF6",
    "Random Forest": "#10B981",
    "Gradient Boosting": "#F59E0B"
}

# Seaborn theme
```

```
plt.figure(figsize=(8, 5))

ax = sns.countplot(
    data=df,
    x="Churn",
    hue="Churn",
    palette={
        0: SUCCESS,
        1: DANGER
    },
    legend=False
)

plt.title(
    "Customer Churn Distribution",
    fontsize=18,
    fontweight="bold",
    pad=20
)

plt.xlabel("Customer Churn", fontsize=12)
plt.ylabel("Number of Customers", fontsize=12)

# Add values on bars
for container in ax.containers:
    ax.bar_label(
```

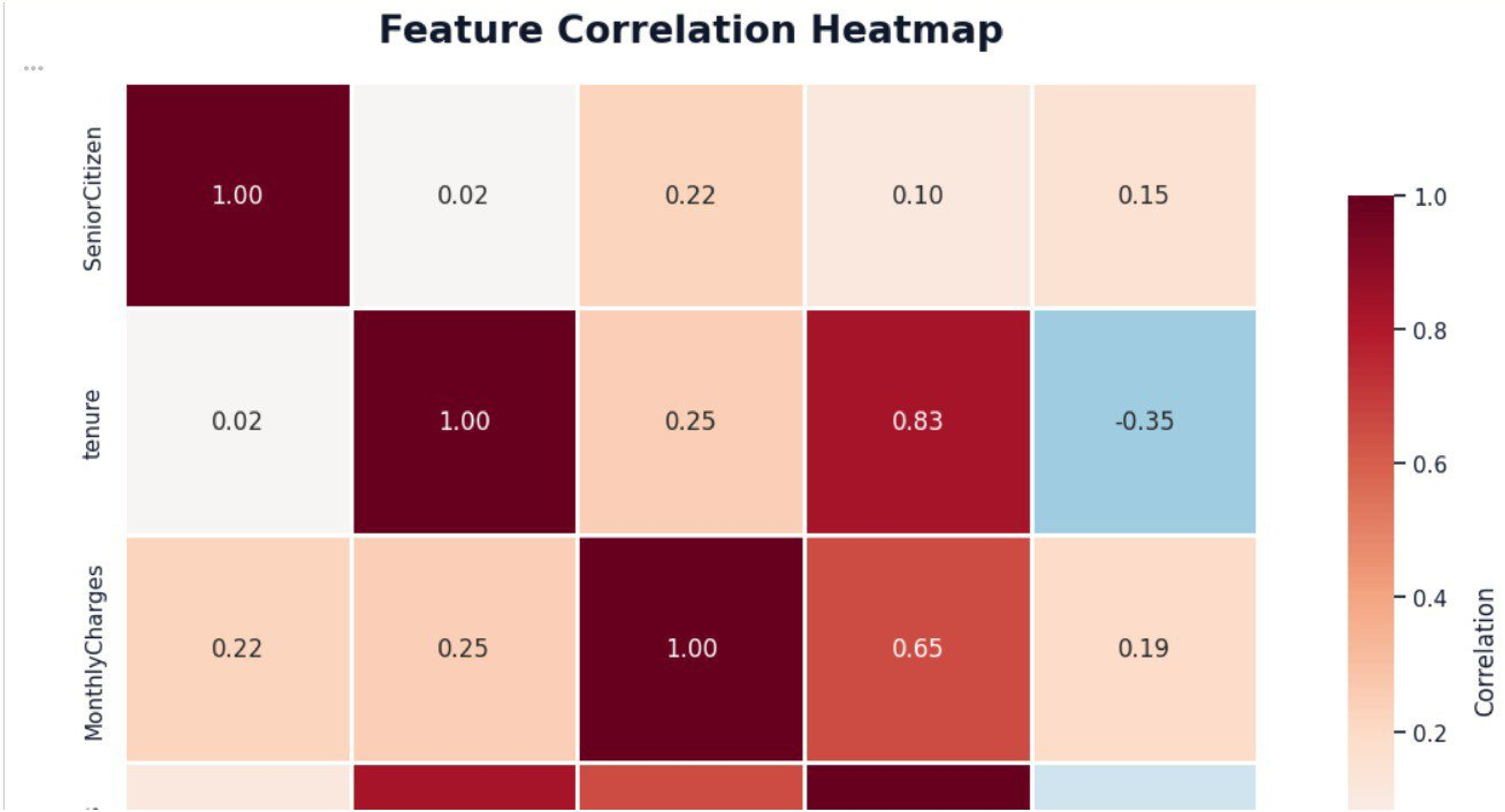

Customer Churn Distribution
5174
1869
Number of Customers
0
1000
2000
3000
4000
5000
0
1
Customer Churn

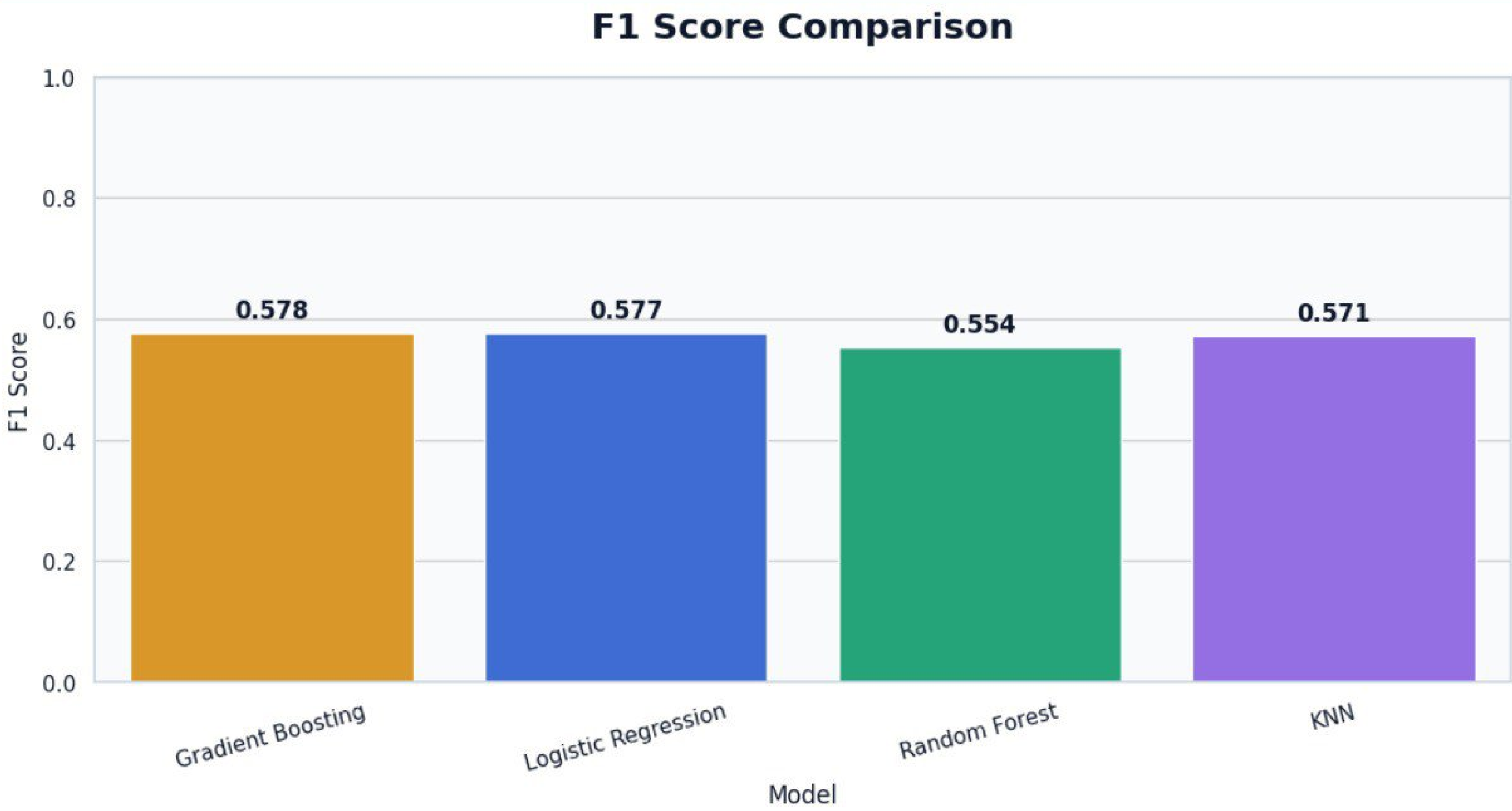
Monthly Charges vs Customer Churn

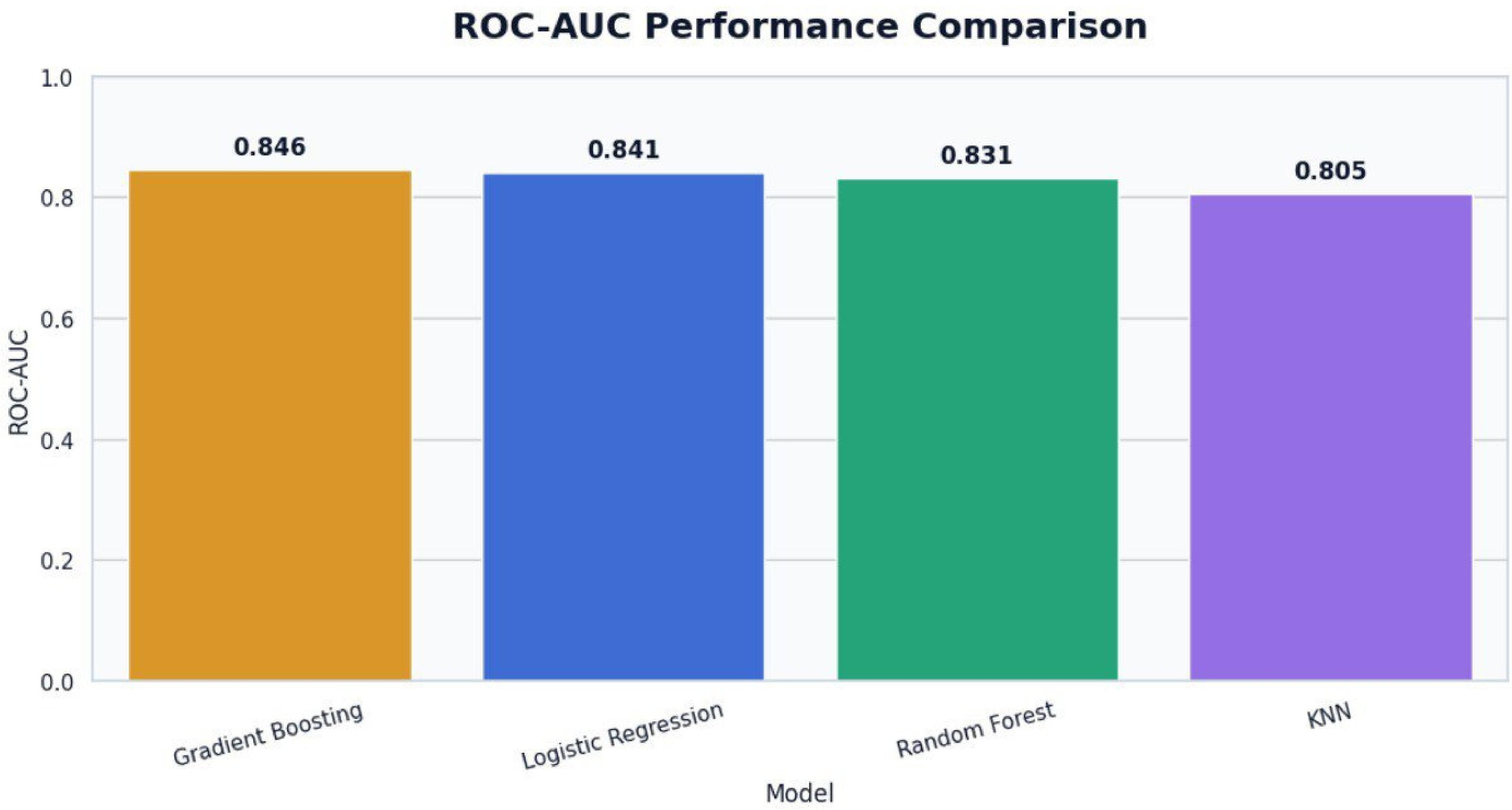

Customer Churn by Contract Type
Churn
No Churn
Churn
2220
1655
1307
166
1647
48
2000
1500
1000
500
0
Number of Customers
Month-to-month
One year
Two year
Contract Type

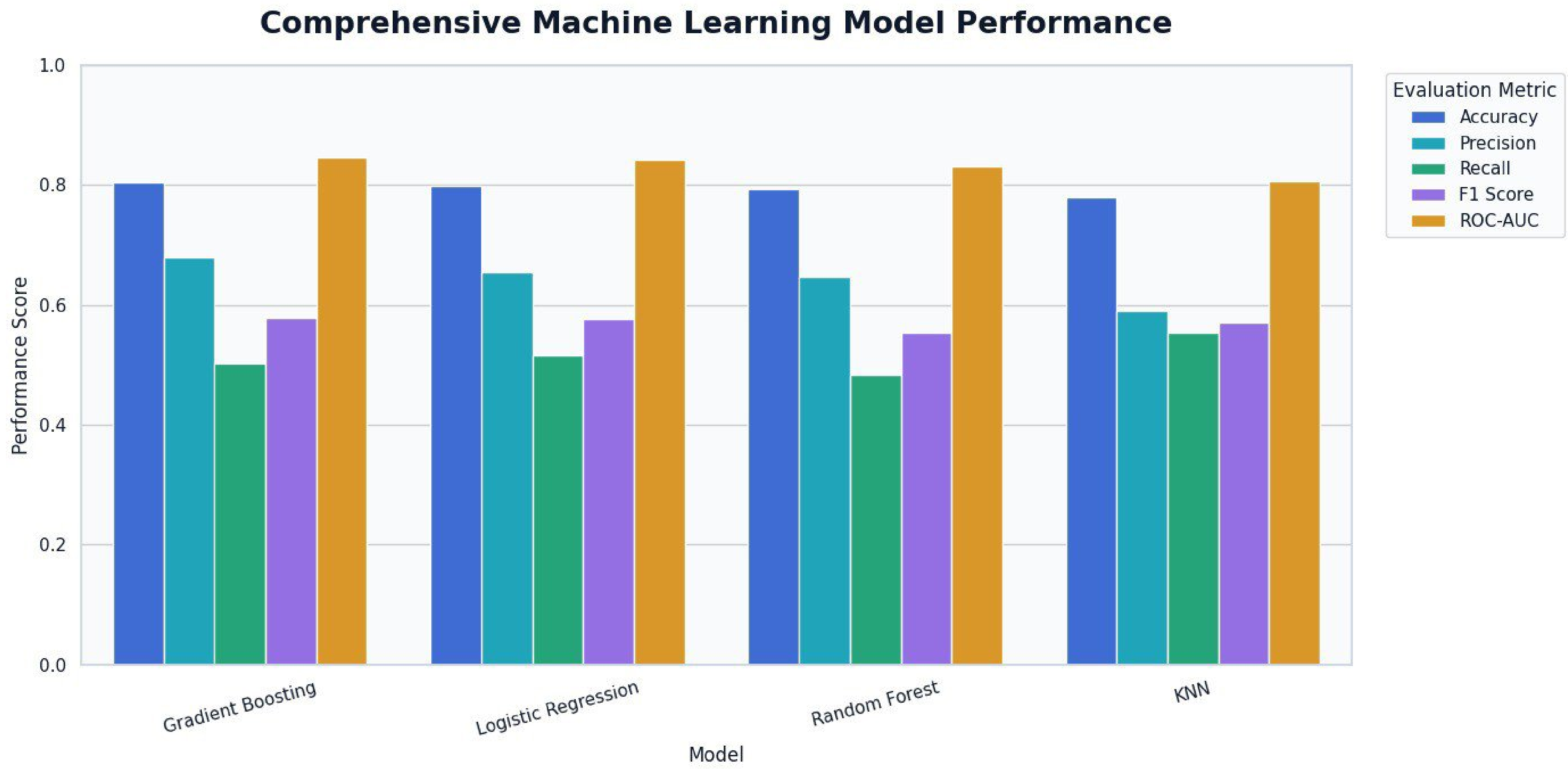


Customer Tenure vs Churn
Churn
0
1
600
500
400
300
200
100
Customer Count


Feature Correlation Heatmap
SeniorCitizen
tenure
MonthlyCharges
1.00
0.02
0.22
0.10
0.15
0.02
1.00
0.25
0.83
-0.35
0.22
0.25
1.00
0.65
0.19
1.0
0.8
0.6
0.4
0.2
Correlation


F1 Score Comparison
1.0
0.8
0.6
0.4
0.2
0.0
F1 Score
0.578
0.577
0.554
0.571
Gradient Boosting
Logistic Regression
Random Forest
KNN
Model

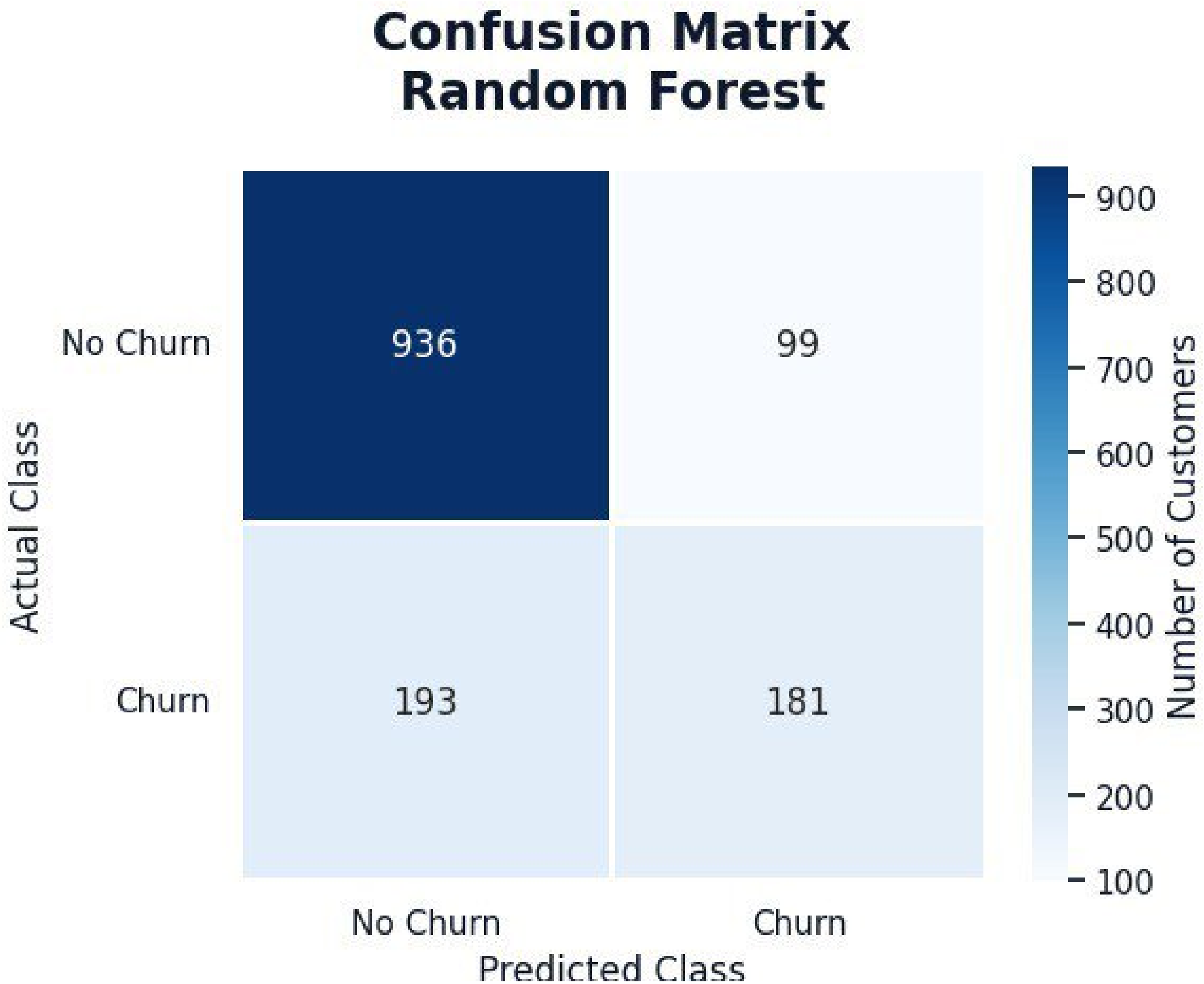

ROC-AUC Performance Comparison
1.0
0.8
0.6
0.4
0.2
0.0
ROC-AUC
0.846
0.841
0.831
0.805
Gradient Boosting
Logistic Regression
Random Forest
KNN
Model

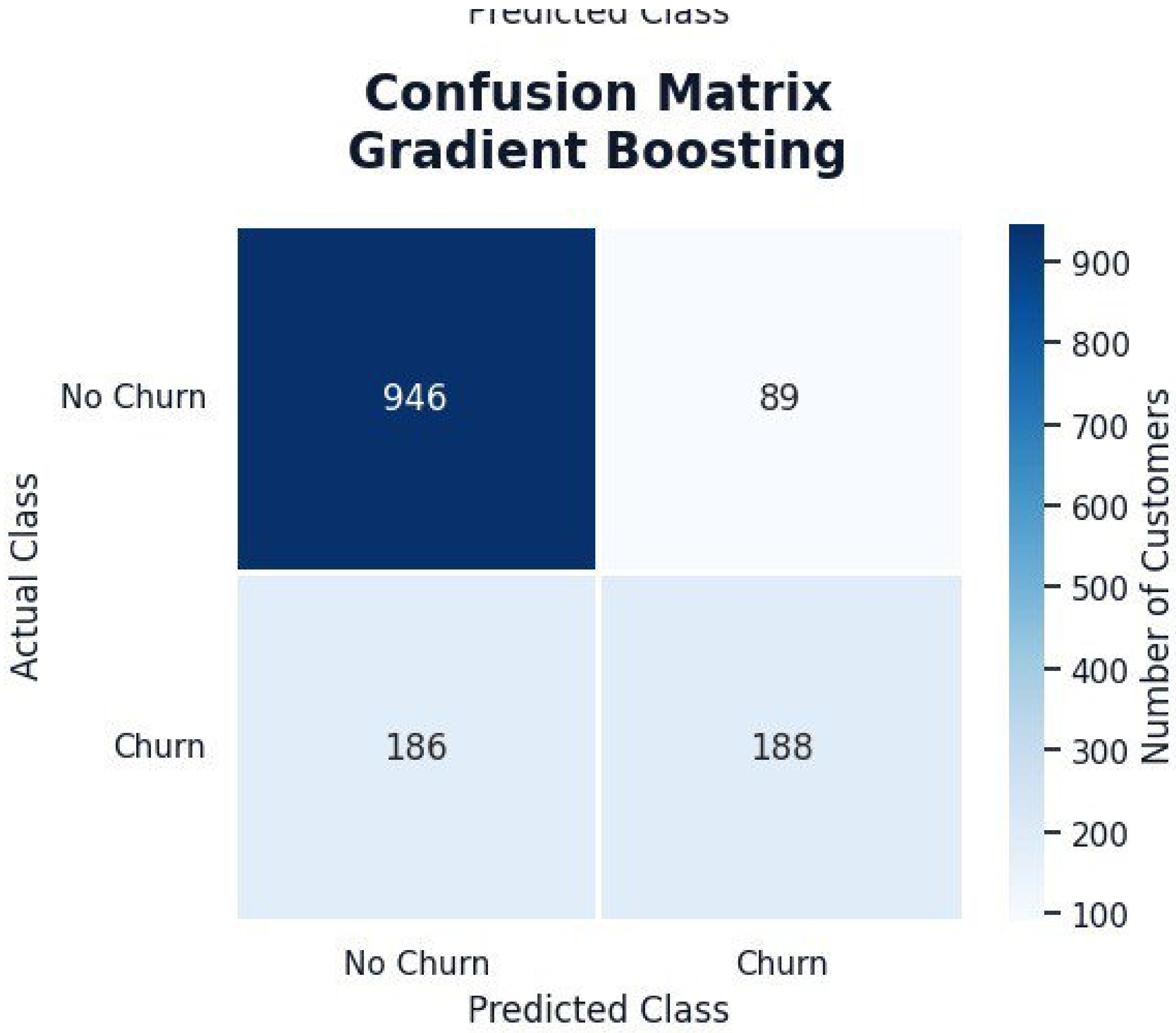


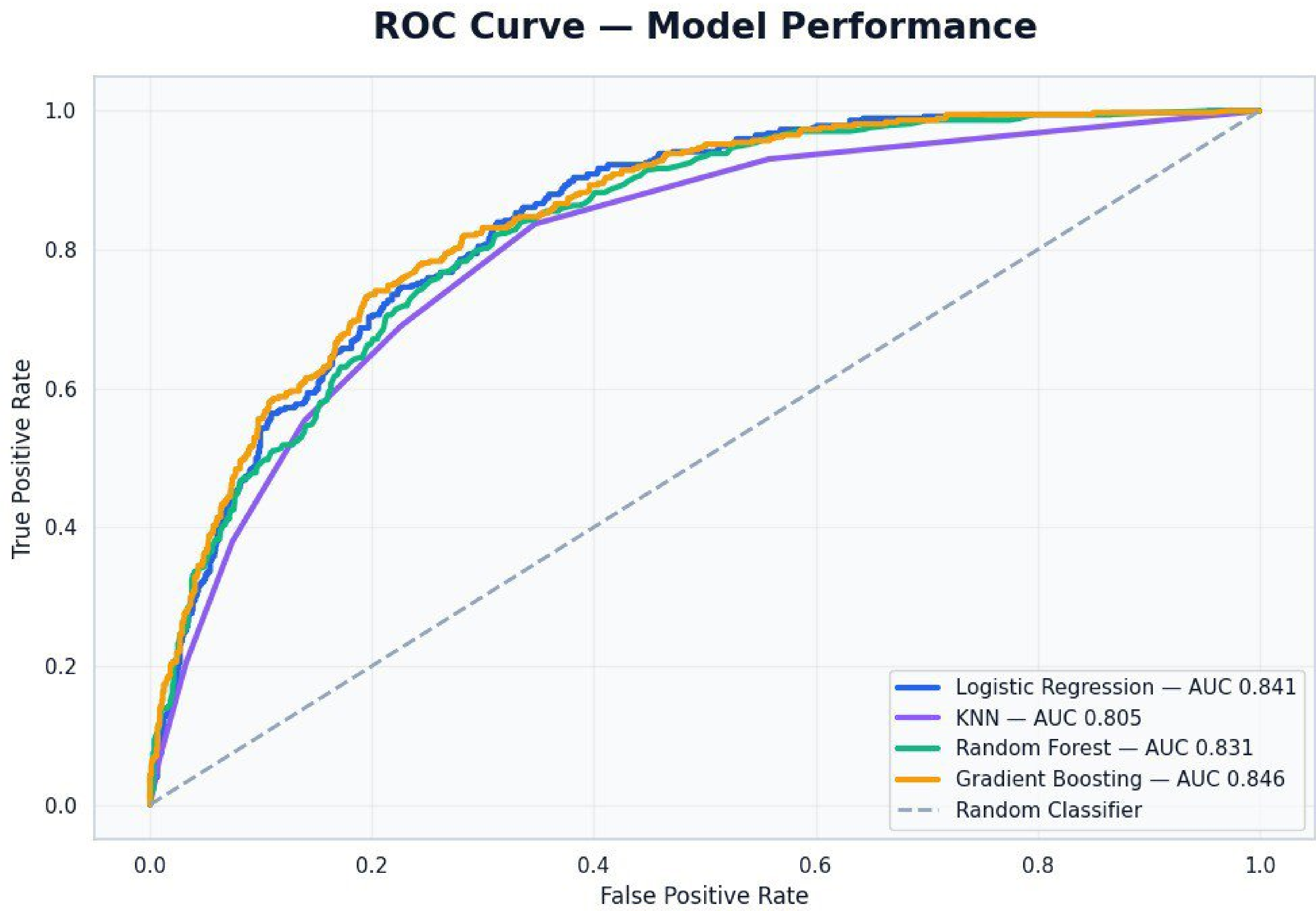
Comprehensive Machine Learning Model Performance
Performance Score
1.0
0.8
0.6
0.4
0.2
0.0
Gradient Boosting
Logistic Regression
Random Forest
KNN
Model
Evaluation Metric
Accuracy
Precision
Recall
F1 Score
ROC-AUC

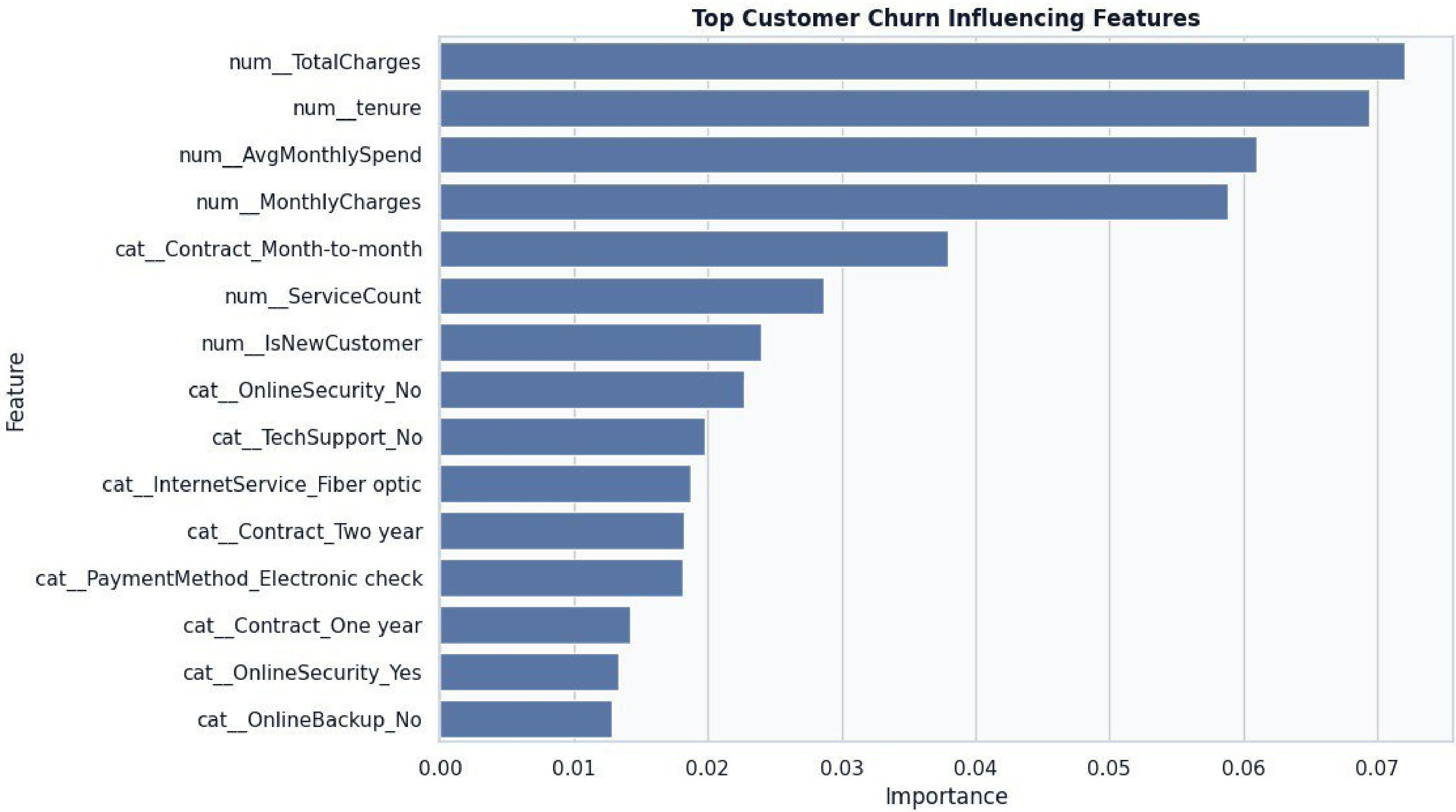
Predicted Class

...




Predicted Class
Confusion Matrix
Gradient Boosting
No Churn
Churn
946
89
186
188
Actual Class
No Churn
Churn
Predicted Class
900
800
700
600
500
400
300
200
100
Number of Customers


ROC Curve — Model Performance
1.0
0.8
0.6
0.4
0.2
0.0
True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate
Logistic Regression — AUC 0.841
KNN — AUC 0.805
Random Forest — AUC 0.831
Gradient Boosting — AUC 0.846
Random Classifier


Top Customer Churn Influencing Features
num__TotalCharges
num__tenure
num__AvgMonthlySpend
num__MonthlyCharges
cat__Contract_Month-to-month
num__ServiceCount
num__IsNewCustomer
cat__OnlineSecurity_No
cat__TechSupport_No
cat__InternetService_Fiber optic
cat__Contract_Two year
cat__PaymentMethod_Electronic check
cat__Contract_One year
cat__OnlineSecurity_Yes
cat__OnlineBackup_No
Feature
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Importance

	Feature	Importance
3	num__TotalCharges	0.072081
1	num__tenure	0.069350
4	num__AvgMonthlySpend	0.060990
2	num__MonthlyCharges	0.058889
5674	cat__Contract_Month-to-month	0.037905
7	num__ServiceCount	0.028696
5	num__IsNewCustomer	0.023943
5656	cat__OnlineSecurity_No	0.022691
5665	cat__TechSupport_No	0.019831
5654	cat__InternetService_Fiber optic	0.018759
5676	cat__Contract_Two year	0.018270
5681	cat__PaymentMethod_Electronic check	0.018135
5675	cat__Contract_One year	0.014197
5658	cat__OnlineSecurity_Yes	0.013355
5659	cat__OnlineBackup_No	0.012847